Appendix E: Detailed Result

Dashboard

Table E1: Dashboard - main threshold results

Summary block extracted from the Dashboard sheet.

Main threshold result for Unitree G1 at $13,500	SF Express	JD Logistics
Normal season: required pph for unit-cost break-even	336.1	242.4
Normal season: required pph for 3-year payback	394.4	284.5
At q=300 pph: max G1 price for 3-year payback	4,112	15,644
At q=400 pph: max G1 price for 3-year payback	14,056	29,432
Conclusion	For SF, G1 must reach ~395 pph or cost <= ~$4.1k at 300 pph for a 3-year payback. At 400 pph, $13.5k becomes acceptable.	For JD, G1 is viable at 300 pph because the 3-year payback price ceiling is ~$15.6k.

Table E2: Dashboard - seasonal G1 required throughput

Seasonal thresholds for the Unitree G1 benchmark.

Brand	Season	Demand/day	Human UC ($/parcel)	q__BE unit	q__BE 3yr	Interpretation
SF Express	Cold/off-peak	50,000	0.0131	363.3	426.2	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.
SF Express	Normal	100,000	0.0152	336.1	394.4	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.
SF Express	Hot/peak	200,000	0.0214	292.4	343.1	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.
JD Logistics	Cold/off-peak	50,000	0.0181	262	307.4	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.
JD Logistics	Normal	100,000	0.0210	242.4	284.5	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.
JD Logistics	Hot/peak	200,000	0.0297	210.9	247.4	Harder in cold/off-peak; easier in peak because labour becomes more expensive relative to robotic capacity.

Parameters

Table E3: Parameters

All monetary values are in USD. The exchange-rate assumption is retained from the earlier model and can be edited.

Parameter	Value	Unit	Role in model	Source / rationale	Column 6
FX__CNY__per__USD	6.78	CNY/USD	Converts salary and cost data	Editable assumption from earlier USD workbook	
Workdays per year	312	days	Annual capacity conversion	26 days/month x 12	
Robot useful life	5	years	Depreciation	Base deployment assumption	
Residual rate	0.20	share of bare price	Depreciation	Base residual assumption	
Target payback	3	years	Investment break-even	Commercial investment threshold	
End-effector upgrade rate	0.10	share of bare price	Upfront robot deployment	Gripper/suction/dexterous-hand adaptation	
Safety system rate	0.0300	share of bare price	Upfront robot deployment	E-stop/protection/safety deployment	
Integration rate	0.20	share of bare price	Upfront robot deployment	WMS/WCS integration and site engineering	
Barcode/OCR system	737.4	USD upfront	Upfront robot deployment	Scanner/OCR proxy	
Maintenance rate	0.0800	share of bare price/year	Annual robot OPEX	Warehouse robotics maintenance proxy	
Annual software	4,247	USD/year	Annual robot OPEX	Robot software/cloud proxy	
Annual operations personnel allocation	3,318	USD/year	Annual robot OPEX	Robotics technician allocation	
Annual electricity	339.8	USD/year	Annual robot OPEX	Charging/electricity proxy	
Annual downtime loss	5,309	USD/year	Annual robot OPEX	Downtime-loss proxy	
Robot cost coefficient a	0.31	USD/year per USD price	C_R(P)=aP+b	Derived from depreciation and maintenance	
Robot fixed annual cost b	13,362	USD/year	C_R(P)=aP+b	Derived from scanner depreciation and fixed OPEX	
Human input	SF Express	JD Logistics	Unit	Source / rationale	
Annual employer cost	14,794	15,629	USD/FTE/year	Derived from uploaded salary component table	
Base human unit cost	0.0152	0.0210	USD/parcel	Cost divided by quality-adjusted annual parcels	
Base annual good parcels	975,834	743,493	parcels/FTE/year	Derived from efficiency workbook	
Base monthly midpoint	948.3	1,000	USD/month	Derived from recruitment salary data	
Season	Demand/day	Human capacity multiplier	Robot capacity multiplier	Labour cost multiplier	Robotable parcel share
Cold/off-peak	50,000	1.10	1	0.95	0.75
Normal	100,000	1	1	1	0.70
Hot/peak	200,000	0.85	0.95	1.20	0.60

Interactive Calculator

Table E4: Interactive_Calculator

Edit the four input cells in column B to test a dynamic robot price-efficiency case.

Input	Value	Unit	Description
Robot bare price P	13,500	USD	Edit this value
Robot throughput q	300	parcels/hour	Edit this value
Carrier	SF Express		Use SF Express or JD Logistics
Season	Normal		Use Cold/off-peak, Normal, or Hot/peak
Output	Formula result	Unit	Decision interpretation
Human unit cost in selected season	0.0152	USD/parcel	
Robot annual output multiplier	3,433	annual parcels per pph	
Robot annual cost	17,493	USD/year	C_R(P)=0.306P+13,362.44

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Input	Value	Unit	Description
Robot annual output	1,029,820	parcels/year	q multiplied by seasonal robot multiplier
Robot unit cost	0.0170	USD/parcel	Lower than human UC means unit-cost efficient
Unit-cost efficient?	No		
Required q for unit-cost break-even	336.1	parcels/hour	Current q must be at least this value
Required q for 3-year payback	394.4	parcels/hour	More conservative investment criterion
Maximum price for unit-cost break-even at current q	7,353	USD	
Maximum price for 3-year payback at current q	4,112	USD	
Lookup table used by calculator			
Season	SF Express Human UC	JD Logistics Human UC	Robot annual multiplier
Cold/off-peak	0.0131	0.0182	3,678
Normal	0.0152	0.0210	3,433
Hot/peak	0.0214	0.0297	2,795

G1 Thresholds

Table E5: G1_Thresholds

This table answers: at Unitree G1 price = $13,500, what throughput is needed?

Brand	Season	Demand/day	Seasonal human UC	q for unit-cost break-even	q for 3-year payback	Main managerial reading	Criterion
SF Express	Cold/off-peak	50,000	0.0131	363.3	426.2	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold
SF Express	Normal	100,000	0.0152	336.1	394.4	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold
SF Express	Hot/peak	200,000	0.0214	292.4	343.1	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold
JD Logistics	Cold/off-peak	50,000	0.0182	262.0	307.4	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold
JD Logistics	Normal	100,000	0.0210	242.4	284.5	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold
JD Logistics	Hot/peak	200,000	0.0297	210.9	247.4	Use the 3-year payback threshold for investment decisions; unit-cost threshold is a lower bar.	Dynamic threshold

Price Efficiency Frontier

Table E6: Price_Efficiency_Frontier - column block 1

For each throughput q, the table reports the maximum bare robot price that can break even.

Brand	Season	Throughput q (pph)	Pmax unit-cost ($)	Pmax 3-year payback ($)	G1 price ($)	G1 feasible by unit cost?	G1 feasible by 3-year payback?
SF Express	Cold/off-peak	200	-12,194	-7,318	13,500	No	No
SF Express	Cold/off-peak	250	-4,326	-2,717	13,500	No	No
SF Express	Cold/off-peak	300	3,543	1,884	13,500	No	No
SF Express	Cold/off-peak	350	11,411	6,485	13,500	No	No
SF Express	Cold/off-peak	400	19,280	11,085	13,500	Yes	No
SF Express	Cold/off-peak	450	27,148	15,686	13,500	Yes	Yes
SF Express	Cold/off-peak	500	35,017	20,287	13,500	Yes	Yes
SF Express	Cold/off-peak	600	50,754	29,489	13,500	Yes	Yes
SF Express	Cold/off-peak	700	66,491	38,690	13,500	Yes	Yes

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Brand	Season	Throughput q (pph)	Pmax unit-cost ($)	Pmax 3-year payback ($)	G1 price ($)	G1 feasible by unit cost?	G1 feasible by 3-year payback?
SF Express	Cold/off-peak	800	82,228	47,892	13,500	Yes	Yes
SF Express	Normal	200	-9,654	-5,833	13,500	No	No
SF Express	Normal	250	-1,151	-860.6	13,500	No	No
SF Express	Normal	300	7,353	4,112	13,500	No	No
SF Express	Normal	350	15,856	9,084	13,500	Yes	No
SF Express	Normal	400	24,360	14,056	13,500	Yes	Yes
SF Express	Normal	450	32,863	19,028	13,500	Yes	Yes
SF Express	Normal	500	41,367	24,000	13,500	Yes	Yes
SF Express	Normal	600	58,374	33,944	13,500	Yes	Yes
SF Express	Normal	700	75,381	43,888	13,500	Yes	Yes
SF Express	Normal	800	92,388	53,833	13,500	Yes	Yes
SF Express	Hot/peak	200	-4,566	-2,858	13,500	No	No
SF Express	Hot/peak	250	5,209	2,858	13,500	No	No
SF Express	Hot/peak	300	14,985	8,574	13,500	Yes	No
SF Express	Hot/peak	350	24,760	14,290	13,500	Yes	Yes
SF Express	Hot/peak	400	34,536	20,006	13,500	Yes	Yes
SF Express	Hot/peak	450	44,311	25,721	13,500	Yes	Yes
SF Express	Hot/peak	500	54,087	31,437	13,500	Yes	Yes
SF Express	Hot/peak	600	73,637	42,869	13,500	Yes	Yes
SF Express	Hot/peak	700	93,188	54,301	13,500	Yes	Yes
SF Express	Hot/peak	800	112,739	65,732	13,500	Yes	Yes
JD Logistics	Cold/off-peak	200	-27.4	-203.9	13,500	No	No
JD Logistics	Cold/off-peak	250	10,883	6,175	13,500	No	No
JD Logistics	Cold/off-peak	300	21,793	12,555	13,500	Yes	No
JD Logistics	Cold/off-peak	350	32,703	18,934	13,500	Yes	Yes
JD Logistics	Cold/off-peak	400	43,613	25,313	13,500	Yes	Yes
JD Logistics	Cold/off-peak	450	54,523	31,693	13,500	Yes	Yes
JD Logistics	Cold/off-peak	500	65,434	38,072	13,500	Yes	Yes
JD Logistics	Cold/off-peak	600	87,254	50,831	13,500	Yes	Yes
JD Logistics	Cold/off-peak	700	109,074	63,589	13,500	Yes	Yes
JD Logistics	Cold/off-peak	800	130,895	76,348	13,500	Yes	Yes
JD Logistics	Normal	200	3,494	1,855	13,500	No	No
JD Logistics	Normal	250	15,285	8,750	13,500	Yes	No
JD Logistics	Normal	300	27,076	15,644	13,500	Yes	Yes
JD Logistics	Normal	350	38,866	22,538	13,500	Yes	Yes
JD Logistics	Normal	400	50,657	29,432	13,500	Yes	Yes
JD Logistics	Normal	450	62,448	36,326	13,500	Yes	Yes
JD Logistics	Normal	500	74,238	43,220	13,500	Yes	Yes
JD Logistics	Normal	600	97,820	57,009	13,500	Yes	Yes
JD Logistics	Normal	700	121,401	70,797	13,500	Yes	Yes
JD Logistics	Normal	800	144,982	84,585	13,500	Yes	Yes
JD Logistics	Hot/peak	200	10,549	5,980	13,500	No	No
JD Logistics	Hot/peak	250	24,103	13,906	13,500	Yes	Yes
JD Logistics	Hot/peak	300	37,658	21,831	13,500	Yes	Yes
JD Logistics	Hot/peak	350	51,212	29,756	13,500	Yes	Yes
JD Logistics	Hot/peak	400	64,766	37,682	13,500	Yes	Yes
JD Logistics	Hot/peak	450	78,320	45,607	13,500	Yes	Yes
JD Logistics	Hot/peak	500	91,875	53,533	13,500	Yes	Yes
JD Logistics	Hot/peak	600	118,983	69,383	13,500	Yes	Yes
JD Logistics	Hot/peak	700	146,092	85,234	13,500	Yes	Yes
JD Logistics	Hot/peak	800	173,200	101,085	13,500	Yes	Yes

Table E7: Price_Efficiency_Frontier - column block 2

Continuation of Price_Efficiency_Frontier; first columns are repeated as row identifiers.

Brand	Season	Throughput q (pph)	Comment
SF Express	Cold/off-peak	200	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	250	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	300	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	350	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	400	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	450	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	500	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	600	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	700	3-year payback is the stricter frontier.
SF Express	Cold/off-peak	800	3-year payback is the stricter frontier.
SF Express	Normal	200	3-year payback is the stricter frontier.
SF Express	Normal	250	3-year payback is the stricter frontier.
SF Express	Normal	300	3-year payback is the stricter frontier.
SF Express	Normal	350	3-year payback is the stricter frontier.
SF Express	Normal	400	3-year payback is the stricter frontier.
SF Express	Normal	450	3-year payback is the stricter frontier.
SF Express	Normal	500	3-year payback is the stricter frontier.
SF Express	Normal	600	3-year payback is the stricter frontier.
SF Express	Normal	700	3-year payback is the stricter frontier.
SF Express	Normal	800	3-year payback is the stricter frontier.
SF Express	Hot/peak	200	3-year payback is the stricter frontier.
SF Express	Hot/peak	250	3-year payback is the stricter frontier.
SF Express	Hot/peak	300	3-year payback is the stricter frontier.
SF Express	Hot/peak	350	3-year payback is the stricter frontier.
SF Express	Hot/peak	400	3-year payback is the stricter frontier.
SF Express	Hot/peak	450	3-year payback is the stricter frontier.
SF Express	Hot/peak	500	3-year payback is the stricter frontier.
SF Express	Hot/peak	600	3-year payback is the stricter frontier.
SF Express	Hot/peak	700	3-year payback is the stricter frontier.
SF Express	Hot/peak	800	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	200	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	250	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	300	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	350	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	400	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	450	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	500	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	600	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	700	3-year payback is the stricter frontier.
JD Logistics	Cold/off-peak	800	3-year payback is the stricter frontier.
JD Logistics	Normal	200	3-year payback is the stricter frontier.
JD Logistics	Normal	250	3-year payback is the stricter frontier.
JD Logistics	Normal	300	3-year payback is the stricter frontier.
JD Logistics	Normal	350	3-year payback is the stricter frontier.

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Brand	Season	Throughput q (pph)	Comment
JD Logistics	Normal	400	3-year payback is the stricter frontier.
JD Logistics	Normal	450	3-year payback is the stricter frontier.
JD Logistics	Normal	500	3-year payback is the stricter frontier.
JD Logistics	Normal	600	3-year payback is the stricter frontier.
JD Logistics	Normal	700	3-year payback is the stricter frontier.
JD Logistics	Normal	800	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	200	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	250	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	300	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	350	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	400	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	450	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	500	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	600	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	700	3-year payback is the stricter frontier.
JD Logistics	Hot/peak	800	3-year payback is the stricter frontier.

Simulation Grid

Table E8: Simulation_Grid - column block 1

Each row solves the cost-minimizing integer mix of robots and human workers under demand and robotable-share constraints.

Brand	Season	Demand/day	Price P ($)	Throughput q (pph)	Robot UC ($/parcel)	Human UC seasonal	Unit-cost feasible?
SF Express	Cold/off-peak	50,000	5,000	250	0.0162	0.0131	No
SF Express	Cold/off-peak	50,000	5,000	300	0.0135	0.0131	No
SF Express	Cold/off-peak	50,000	5,000	350	0.0116	0.0131	Yes
SF Express	Cold/off-peak	50,000	5,000	400	0.0101	0.0131	Yes
SF Express	Cold/off-peak	50,000	5,000	500	0.0081	0.0131	Yes
SF Express	Cold/off-peak	50,000	5,000	600	0.0067	0.0131	Yes
SF Express	Cold/off-peak	50,000	5,000	800	0.0051	0.0131	Yes
SF Express	Cold/off-peak	50,000	10,000	250	0.0179	0.0131	No
SF Express	Cold/off-peak	50,000	10,000	300	0.0149	0.0131	No
SF Express	Cold/off-peak	50,000	10,000	350	0.0128	0.0131	Yes
SF Express	Cold/off-peak	50,000	10,000	400	0.0112	0.0131	Yes
SF Express	Cold/off-peak	50,000	10,000	500	0.0089	0.0131	Yes
SF Express	Cold/off-peak	50,000	10,000	600	0.0074	0.0131	Yes
SF Express	Cold/off-peak	50,000	10,000	800	0.0056	0.0131	Yes
SF Express	Cold/off-peak	50,000	13,500	250	0.0190	0.0131	No
SF Express	Cold/off-peak	50,000	13,500	300	0.0159	0.0131	No
SF Express	Cold/off-peak	50,000	13,500	350	0.0136	0.0131	No
SF Express	Cold/off-peak	50,000	13,500	400	0.0119	0.0131	Yes
SF Express	Cold/off-peak	50,000	13,500	500	0.0095	0.0131	Yes
SF Express	Cold/off-peak	50,000	13,500	600	0.0079	0.0131	Yes
SF Express	Cold/off-peak	50,000	13,500	800	0.0059	0.0131	Yes
SF Express	Cold/off-peak	50,000	20,000	250	0.0212	0.0131	No
SF Express	Cold/off-peak	50,000	20,000	300	0.0177	0.0131	No
SF Express	Cold/off-peak	50,000	20,000	350	0.0151	0.0131	No
SF Express	Cold/off-peak	50,000	20,000	400	0.0132	0.0131	No
SF Express	Cold/off-peak	50,000	20,000	500	0.0106	0.0131	Yes
SF Express	Cold/off-peak	50,000	20,000	600	0.0088	0.0131	Yes
SF Express	Cold/off-peak	50,000	20,000	800	0.0066	0.0131	Yes
SF Express	Cold/off-peak	50,000	30,000	250	0.0245	0.0131	No
SF Express	Cold/off-peak	50,000	30,000	300	0.0204	0.0131	No
SF Express	Cold/off-peak	50,000	30,000	350	0.0175	0.0131	No
SF Express	Cold/off-peak	50,000	30,000	400	0.0153	0.0131	No
SF Express	Cold/off-peak	50,000	30,000	500	0.0123	0.0131	Yes
SF Express	Cold/off-peak	50,000	30,000	600	0.0102	0.0131	Yes
SF Express	Cold/off-peak	50,000	30,000	800	0.0077	0.0131	Yes
SF Express	Cold/off-peak	50,000	50,000	250	0.0312	0.0131	No
SF Express	Cold/off-peak	50,000	50,000	300	0.0260	0.0131	No
SF Express	Cold/off-peak	50,000	50,000	350	0.0223	0.0131	No
SF Express	Cold/off-peak	50,000	50,000	400	0.0195	0.0131	No
SF Express	Cold/off-peak	50,000	50,000	500	0.0156	0.0131	No
SF Express	Cold/off-peak	50,000	50,000	600	0.0130	0.0131	Yes
SF Express	Cold/off-peak	50,000	50,000	800	0.0097	0.0131	Yes
SF Express	Normal	100,000	5,000	250	0.0174	0.0152	No
SF Express	Normal	100,000	5,000	300	0.0145	0.0152	Yes
SF Express	Normal	100,000	5,000	350	0.0124	0.0152	Yes
SF Express	Normal	100,000	5,000	400	0.0108	0.0152	Yes
SF Express	Normal	100,000	5,000	500	0.0087	0.0152	Yes
SF Express	Normal	100,000	5,000	600	0.0072	0.0152	Yes
SF Express	Normal	100,000	5,000	800	0.0054	0.0152	Yes
SF Express	Normal	100,000	10,000	250	0.0191	0.0152	No
SF Express	Normal	100,000	10,000	300	0.0159	0.0152	No
SF Express	Normal	100,000	10,000	350	0.0137	0.0152	Yes
SF Express	Normal	100,000	10,000	400	0.0120	0.0152	Yes
SF Express	Normal	100,000	10,000	500	0.0096	0.0152	Yes
SF Express	Normal	100,000	10,000	600	0.0080	0.0152	Yes
SF Express	Normal	100,000	10,000	800	0.0060	0.0152	Yes
SF Express	Normal	100,000	13,500	250	0.0204	0.0152	No
SF Express	Normal	100,000	13,500	300	0.0170	0.0152	No
SF Express	Normal	100,000	13,500	350	0.0146	0.0152	Yes
SF Express	Normal	100,000	13,500	400	0.0127	0.0152	Yes
SF Express	Normal	100,000	13,500	500	0.0102	0.0152	Yes
SF Express	Normal	100,000	13,500	600	0.0085	0.0152	Yes
SF Express	Normal	100,000	13,500	800	0.0064	0.0152	Yes
SF Express	Normal	100,000	20,000	250	0.0227	0.0152	No
SF Express	Normal	100,000	20,000	300	0.0189	0.0152	No
SF Express	Normal	100,000	20,000	350	0.0162	0.0152	No
SF Express	Normal	100,000	20,000	400	0.0142	0.0152	Yes
SF Express	Normal	100,000	20,000	500	0.0114	0.0152	Yes
SF Express	Normal	100,000	20,000	600	0.0095	0.0152	Yes
SF Express	Normal	100,000	20,000	800	0.0071	0.0152	Yes
SF Express	Normal	100,000	30,000	250	0.0263	0.0152	No
SF Express	Normal	100,000	30,000	300	0.0219	0.0152	No
SF Express	Normal	100,000	30,000	350	0.0188	0.0152	No
SF Express	Normal	100,000	30,000	400	0.0164	0.0152	No
SF Express	Normal	100,000	30,000	500	0.0131	0.0152	Yes
SF Express	Normal	100,000	30,000	600	0.0109	0.0152	Yes
SF Express	Normal	100,000	30,000	800	0.0082	0.0152	Yes
SF Express	Normal	100,000	50,000	250	0.0334	0.0152	No
SF Express	Normal	100,000	50,000	300	0.0278	0.0152	No
SF Express	Normal	100,000	50,000	350	0.0239	0.0152	No
SF Express	Normal	100,000	50,000	400	0.0209	0.0152	No
SF Express	Normal	100,000	50,000	500	0.0167	0.0152	No
SF Express	Normal	100,000	50,000	600	0.0139	0.0152	Yes
SF Express	Normal	100,000	50,000	800	0.0104	0.0152	Yes
SF Express	Hot/peak	200,000	5,000	250	0.0213	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	300	0.0178	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	350	0.0152	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	400	0.0133	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	500	0.0107	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	600	0.0089	0.0214	Yes
SF Express	Hot/peak	200,000	5,000	800	0.0067	0.0214	Yes
SF Express	Hot/peak	200,000	10,000	250	0.0235	0.0214	No
SF Express	Hot/peak	200,000	10,000	300	0.0196	0.0214	Yes
SF Express	Hot/peak	200,000	10,000	350	0.0168	0.0214	Yes

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Brand	Season	Demand/day	Price P ($)	Throughput q (pph)	Robot UC ($/parcel)	Human UC seasonal	Unit-cost feasible?
SF Express	Hot/peak	200,000	10,000	400	0.0147	0.0214	Yes
SF Express	Hot/peak	200,000	10,000	500	0.0118	0.0214	Yes
SF Express	Hot/peak	200,000	10,000	600	0.0098	0.0214	Yes
SF Express	Hot/peak	200,000	10,000	800	0.0073	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	250	0.0250	0.0214	No
SF Express	Hot/peak	200,000	13,500	300	0.0209	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	350	0.0179	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	400	0.0156	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	500	0.0125	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	600	0.0104	0.0214	Yes
SF Express	Hot/peak	200,000	13,500	800	0.0078	0.0214	Yes
SF Express	Hot/peak	200,000	20,000	250	0.0279	0.0214	No
SF Express	Hot/peak	200,000	20,000	300	0.0232	0.0214	No
SF Express	Hot/peak	200,000	20,000	350	0.0199	0.0214	Yes
SF Express	Hot/peak	200,000	20,000	400	0.0174	0.0214	Yes
SF Express	Hot/peak	200,000	20,000	500	0.0139	0.0214	Yes
SF Express	Hot/peak	200,000	20,000	600	0.0116	0.0214	Yes
SF Express	Hot/peak	200,000	20,000	800	0.0087	0.0214	Yes
SF Express	Hot/peak	200,000	30,000	250	0.0323	0.0214	No
SF Express	Hot/peak	200,000	30,000	300	0.0269	0.0214	No
SF Express	Hot/peak	200,000	30,000	350	0.0230	0.0214	No
SF Express	Hot/peak	200,000	30,000	400	0.0202	0.0214	Yes
SF Express	Hot/peak	200,000	30,000	500	0.0161	0.0214	Yes
SF Express	Hot/peak	200,000	30,000	600	0.0134	0.0214	Yes
SF Express	Hot/peak	200,000	30,000	800	0.0101	0.0214	Yes
SF Express	Hot/peak	200,000	50,000	250	0.0410	0.0214	No
SF Express	Hot/peak	200,000	50,000	300	0.0342	0.0214	No
SF Express	Hot/peak	200,000	50,000	350	0.0293	0.0214	No
SF Express	Hot/peak	200,000	50,000	400	0.0256	0.0214	No
SF Express	Hot/peak	200,000	50,000	500	0.0205	0.0214	Yes
SF Express	Hot/peak	200,000	50,000	600	0.0171	0.0214	Yes
SF Express	Hot/peak	200,000	50,000	800	0.0128	0.0214	Yes
JD Logistics	Cold/off-peak	50,000	5,000	250	0.0162	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	300	0.0135	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	350	0.0116	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	400	0.0101	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	500	0.0081	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	600	0.0067	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	5,000	800	0.0051	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	250	0.0179	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	300	0.0149	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	350	0.0128	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	400	0.0112	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	500	0.0089	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	600	0.0074	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	10,000	800	0.0056	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	250	0.0190	0.0182	No
JD Logistics	Cold/off-peak	50,000	13,500	300	0.0159	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	350	0.0136	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	400	0.0119	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	500	0.0095	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	600	0.0079	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	13,500	800	0.0059	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	250	0.0212	0.0182	No
JD Logistics	Cold/off-peak	50,000	20,000	300	0.0177	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	350	0.0151	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	400	0.0132	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	500	0.0106	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	600	0.0088	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	20,000	800	0.0066	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	30,000	250	0.0245	0.0182	No
JD Logistics	Cold/off-peak	50,000	30,000	300	0.0204	0.0182	No
JD Logistics	Cold/off-peak	50,000	30,000	350	0.0175	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	30,000	400	0.0153	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	30,000	500	0.0123	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	30,000	600	0.0102	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	30,000	800	0.0077	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	50,000	250	0.0312	0.0182	No
JD Logistics	Cold/off-peak	50,000	50,000	300	0.0260	0.0182	No
JD Logistics	Cold/off-peak	50,000	50,000	350	0.0223	0.0182	No
JD Logistics	Cold/off-peak	50,000	50,000	400	0.0195	0.0182	No
JD Logistics	Cold/off-peak	50,000	50,000	500	0.0156	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	50,000	600	0.0130	0.0182	Yes
JD Logistics	Cold/off-peak	50,000	50,000	800	0.0097	0.0182	Yes
JD Logistics	Normal	100,000	5,000	250	0.0174	0.0210	Yes
JD Logistics	Normal	100,000	5,000	300	0.0145	0.0210	Yes
JD Logistics	Normal	100,000	5,000	350	0.0124	0.0210	Yes
JD Logistics	Normal	100,000	5,000	400	0.0108	0.0210	Yes
JD Logistics	Normal	100,000	5,000	500	0.0087	0.0210	Yes
JD Logistics	Normal	100,000	5,000	600	0.0072	0.0210	Yes
JD Logistics	Normal	100,000	5,000	800	0.0054	0.0210	Yes
JD Logistics	Normal	100,000	10,000	250	0.0191	0.0210	Yes
JD Logistics	Normal	100,000	10,000	300	0.0159	0.0210	Yes
JD Logistics	Normal	100,000	10,000	350	0.0137	0.0210	Yes
JD Logistics	Normal	100,000	10,000	400	0.0120	0.0210	Yes
JD Logistics	Normal	100,000	10,000	500	0.0096	0.0210	Yes
JD Logistics	Normal	100,000	10,000	600	0.0080	0.0210	Yes
JD Logistics	Normal	100,000	10,000	800	0.0060	0.0210	Yes
JD Logistics	Normal	100,000	13,500	250	0.0204	0.0210	Yes
JD Logistics	Normal	100,000	13,500	300	0.0170	0.0210	Yes
JD Logistics	Normal	100,000	13,500	350	0.0146	0.0210	Yes
JD Logistics	Normal	100,000	13,500	400	0.0127	0.0210	Yes
JD Logistics	Normal	100,000	13,500	500	0.0102	0.0210	Yes
JD Logistics	Normal	100,000	13,500	600	0.0085	0.0210	Yes
JD Logistics	Normal	100,000	13,500	800	0.0064	0.0210	Yes
JD Logistics	Normal	100,000	20,000	250	0.0227	0.0210	No
JD Logistics	Normal	100,000	20,000	300	0.0189	0.0210	Yes
JD Logistics	Normal	100,000	20,000	350	0.0162	0.0210	Yes
JD Logistics	Normal	100,000	20,000	400	0.0142	0.0210	Yes
JD Logistics	Normal	100,000	20,000	500	0.0114	0.0210	Yes

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Brand	Season	Demand/day	Price P ($)	Throughput q (pph)	Robot UC ($/parcel)	Human UC seasonal	Unit-cost feasible?
JD Logistics	Normal	100,000	20,000	600	0.0095	0.0210	Yes
JD Logistics	Normal	100,000	20,000	800	0.0071	0.0210	Yes
JD Logistics	Normal	100,000	30,000	250	0.0263	0.0210	No
JD Logistics	Normal	100,000	30,000	300	0.0219	0.0210	No
JD Logistics	Normal	100,000	30,000	350	0.0188	0.0210	Yes
JD Logistics	Normal	100,000	30,000	400	0.0164	0.0210	Yes
JD Logistics	Normal	100,000	30,000	500	0.0131	0.0210	Yes
JD Logistics	Normal	100,000	30,000	600	0.0109	0.0210	Yes
JD Logistics	Normal	100,000	30,000	800	0.0082	0.0210	Yes
JD Logistics	Normal	100,000	50,000	250	0.0334	0.0210	No
JD Logistics	Normal	100,000	50,000	300	0.0278	0.0210	No
JD Logistics	Normal	100,000	50,000	350	0.0239	0.0210	No
JD Logistics	Normal	100,000	50,000	400	0.0209	0.0210	Yes
JD Logistics	Normal	100,000	50,000	500	0.0167	0.0210	Yes
JD Logistics	Normal	100,000	50,000	600	0.0139	0.0210	Yes
JD Logistics	Normal	100,000	50,000	800	0.0104	0.0210	Yes
JD Logistics	Hot/peak	200,000	5,000	250	0.0213	0.0297	Yes
JD Logistics	Hot/peak	200,000	5,000	300	0.0178	0.0297	Yes
JD Logistics	Hot/peak	200,000	5,000	350	0.0152	0.0297	Yes
JD Logistics	Hot/peak	200,000	5,000	400	0.0133	0.0297	Yes
JD Logistics	Hot/peak	200,000	5,000	500	0.0107	0.0297	Yes
JD Logistics	Hot/peak	200,000	5,000	600	0.0089	0.0297	Yes
JD Logistics	Hot/peak	200,000	5,000	800	0.0067	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	250	0.0235	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	300	0.0196	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	350	0.0168	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	400	0.0147	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	500	0.0118	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	600	0.0098	0.0297	Yes
JD Logistics	Hot/peak	200,000	10,000	800	0.0073	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	250	0.0250	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	300	0.0209	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	350	0.0179	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	400	0.0156	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	500	0.0125	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	600	0.0104	0.0297	Yes
JD Logistics	Hot/peak	200,000	13,500	800	0.0078	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	250	0.0279	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	300	0.0232	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	350	0.0199	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	400	0.0174	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	500	0.0139	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	600	0.0116	0.0297	Yes
JD Logistics	Hot/peak	200,000	20,000	800	0.0087	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	250	0.0323	0.0297	No
JD Logistics	Hot/peak	200,000	30,000	300	0.0269	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	350	0.0230	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	400	0.0202	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	500	0.0161	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	600	0.0134	0.0297	Yes
JD Logistics	Hot/peak	200,000	30,000	800	0.0101	0.0297	Yes
JD Logistics	Hot/peak	200,000	50,000	250	0.0410	0.0297	No
JD Logistics	Hot/peak	200,000	50,000	300	0.0342	0.0297	No
JD Logistics	Hot/peak	200,000	50,000	350	0.0293	0.0297	Yes
JD Logistics	Hot/peak	200,000	50,000	400	0.0256	0.0297	Yes
JD Logistics	Hot/peak	200,000	50,000	500	0.0205	0.0297	Yes
JD Logistics	Hot/peak	200,000	50,000	600	0.0171	0.0297	Yes
JD Logistics	Hot/peak	200,000	50,000	800	0.0128	0.0297	Yes

Table E9: Simulation_Grid - column block 2

Continuation of Simulation_Grid; first columns are repeated as row identifiers.

Brand	Season	Demand/day	3-year payback feasible?	Optimal robots	Optimal humans	Robot share	Daily cost ($)
SF Express	Cold/off-peak	50,000	No	4	10	0.31	641.4
SF Express	Cold/off-peak	50,000	No	7	5	0.66	559.4
SF Express	Cold/off-peak	50,000	Yes	6	5	0.66	511.6
SF Express	Cold/off-peak	50,000	Yes	6	4	0.75	466.6
SF Express	Cold/off-peak	50,000	Yes	5	4	0.75	418.8
SF Express	Cold/off-peak	50,000	Yes	4	4	0.75	371.1
SF Express	Cold/off-peak	50,000	Yes	3	4	0.75	323.4
SF Express	Cold/off-peak	50,000	No	4	10	0.31	661.0
SF Express	Cold/off-peak	50,000	No	7	5	0.66	593.7
SF Express	Cold/off-peak	50,000	No	6	5	0.66	541.0
SF Express	Cold/off-peak	50,000	Yes	6	4	0.75	496.0
SF Express	Cold/off-peak	50,000	Yes	5	4	0.75	443.4
SF Express	Cold/off-peak	50,000	Yes	4	4	0.75	390.7
SF Express	Cold/off-peak	50,000	Yes	3	4	0.75	338.1
SF Express	Cold/off-peak	50,000	No	4	10	0.31	674.7
SF Express	Cold/off-peak	50,000	No	7	5	0.66	617.7
SF Express	Cold/off-peak	50,000	No	6	5	0.66	561.6
SF Express	Cold/off-peak	50,000	No	6	4	0.75	516.6
SF Express	Cold/off-peak	50,000	Yes	5	4	0.75	460.5
SF Express	Cold/off-peak	50,000	Yes	4	4	0.75	404.5
SF Express	Cold/off-peak	50,000	Yes	3	4	0.75	348.4
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	7	5	0.66	662.3
SF Express	Cold/off-peak	50,000	No	6	5	0.66	599.9
SF Express	Cold/off-peak	50,000	No	6	4	0.75	554.8
SF Express	Cold/off-peak	50,000	Yes	5	4	0.75	492.4
SF Express	Cold/off-peak	50,000	Yes	4	4	0.75	430.0
SF Express	Cold/off-peak	50,000	Yes	3	4	0.75	367.5
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	1	13	0.11	657.8
SF Express	Cold/off-peak	50,000	No	6	4	0.75	613.7
SF Express	Cold/off-peak	50,000	No	5	4	0.75	541.4
SF Express	Cold/off-peak	50,000	No	4	4	0.75	469.2
SF Express	Cold/off-peak	50,000	Yes	3	4	0.75	396.9
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	0	15	0	675.7
SF Express	Cold/off-peak	50,000	No	2	10	0.31	634.2
SF Express	Cold/off-peak	50,000	No	4	4	0.75	547.7
SF Express	Cold/off-peak	50,000	No	3	4	0.75	455.8
SF Express	Normal	100,000	No	16	12	0.63	1,333
SF Express	Normal	100,000	No	14	11	0.66	1,190
SF Express	Normal	100,000	Yes	12	11	0.66	1,094
SF Express	Normal	100,000	Yes	11	10	0.69	999.2
SF Express	Normal	100,000	Yes	9	10	0.70	903.8
SF Express	Normal	100,000	Yes	7	11	0.66	855.7
SF Express	Normal	100,000	Yes	6	10	0.70	760.6
SF Express	Normal	100,000	No	16	12	0.63	1,411
SF Express	Normal	100,000	No	14	11	0.66	1,258
SF Express	Normal	100,000	No	12	11	0.66	1,153
SF Express	Normal	100,000	Yes	11	10	0.69	1,053
SF Express	Normal	100,000	Yes	9	10	0.70	947.9
SF Express	Normal	100,000	Yes	7	11	0.66	890.0
SF Express	Normal	100,000	Yes	6	10	0.70	790.0
SF Express	Normal	100,000	No	16	12	0.63	1,466
SF Express	Normal	100,000	No	14	11	0.66	1,307
SF Express	Normal	100,000	No	12	11	0.66	1,194
SF Express	Normal	100,000	Yes	11	10	0.69	1,091
SF Express	Normal	100,000	Yes	9	10	0.70	978.8
SF Express	Normal	100,000	Yes	7	11	0.66	914.1
SF Express	Normal	100,000	Yes	6	10	0.70	810.6
SF Express	Normal	100,000	No	0	32	0	1,517
SF Express	Normal	100,000	No	14	11	0.66	1,396
SF Express	Normal	100,000	No	12	11	0.66	1,271
SF Express	Normal	100,000	No	11	10	0.69	1,161
SF Express	Normal	100,000	Yes	9	10	0.70	1,036
SF Express	Normal	100,000	Yes	7	11	0.66	958.7
SF Express	Normal	100,000	Yes	6	10	0.70	848.8
SF Express	Normal	100,000	No	0	32	0	1,517
SF Express	Normal	100,000	No	0	32	0	1,517

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Brand	Season	Demand/day	3-year payback feasible?	Optimal robots	Optimal humans	Robot share	Daily cost ($)
SF Express	Normal	100,000	No	12	11	0.66	1,389
SF Express	Normal	100,000	No	11	10	0.69	1,269
SF Express	Normal	100,000	No	9	10	0.70	1,124
SF Express	Normal	100,000	Yes	7	11	0.66	1,027
SF Express	Normal	100,000	Yes	6	10	0.70	907.7
SF Express	Normal	100,000	No	0	32	0	1,517
SF Express	Normal	100,000	No	0	32	0	1,517
SF Express	Normal	100,000	No	0	32	0	1,517
SF Express	Normal	100,000	No	11	10	0.69	1,485
SF Express	Normal	100,000	No	9	10	0.70	1,301
SF Express	Normal	100,000	No	7	11	0.66	1,165
SF Express	Normal	100,000	Yes	6	10	0.70	1,025
SF Express	Hot/peak	200,000	No	32	31	0.60	3,291
SF Express	Hot/peak	200,000	Yes	27	31	0.60	3,053
SF Express	Hot/peak	200,000	Yes	23	31	0.60	2,862
SF Express	Hot/peak	200,000	Yes	20	31	0.60	2,719
SF Express	Hot/peak	200,000	Yes	16	31	0.60	2,528
SF Express	Hot/peak	200,000	Yes	14	31	0.60	2,432
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,241
SF Express	Hot/peak	200,000	No	32	31	0.60	3,448
SF Express	Hot/peak	200,000	No	27	31	0.60	3,185
SF Express	Hot/peak	200,000	Yes	23	31	0.60	2,975
SF Express	Hot/peak	200,000	Yes	20	31	0.60	2,817
SF Express	Hot/peak	200,000	Yes	16	31	0.60	2,606
SF Express	Hot/peak	200,000	Yes	14	31	0.60	2,501
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,290
SF Express	Hot/peak	200,000	No	32	31	0.60	3,558
SF Express	Hot/peak	200,000	No	27	31	0.60	3,278
SF Express	Hot/peak	200,000	Yes	23	31	0.60	3,053
SF Express	Hot/peak	200,000	Yes	20	31	0.60	2,885
SF Express	Hot/peak	200,000	Yes	16	31	0.60	2,661
SF Express	Hot/peak	200,000	Yes	14	31	0.60	2,549
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,325
SF Express	Hot/peak	200,000	No	31	32	0.58	3,757
SF Express	Hot/peak	200,000	No	26	32	0.58	3,444
SF Express	Hot/peak	200,000	No	22	32	0.57	3,195
SF Express	Hot/peak	200,000	Yes	20	31	0.60	3,013
SF Express	Hot/peak	200,000	Yes	16	31	0.60	2,763
SF Express	Hot/peak	200,000	Yes	13	32	0.58	2,633
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,388
SF Express	Hot/peak	200,000	No	31	32	0.58	4,061
SF Express	Hot/peak	200,000	No	26	32	0.58	3,699
SF Express	Hot/peak	200,000	No	22	32	0.57	3,410
SF Express	Hot/peak	200,000	No	20	31	0.60	3,209
SF Express	Hot/peak	200,000	Yes	16	31	0.60	2,920
SF Express	Hot/peak	200,000	Yes	13	32	0.58	2,760
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,486
SF Express	Hot/peak	200,000	No	1	74	0.0187	4,302
SF Express	Hot/peak	200,000	No	24	35	0.54	4,196
SF Express	Hot/peak	200,000	No	22	32	0.57	3,842
SF Express	Hot/peak	200,000	No	20	31	0.60	3,601
SF Express	Hot/peak	200,000	No	16	31	0.60	3,234
SF Express	Hot/peak	200,000	No	13	32	0.58	3,015
SF Express	Hot/peak	200,000	Yes	10	31	0.60	2,683
JD Logistics	Cold/off-peak	50,000	Yes	9	6	0.71	715.1
JD Logistics	Cold/off-peak	50,000	Yes	8	5	0.75	619.8
JD Logistics	Cold/off-peak	50,000	Yes	7	5	0.75	572.1
JD Logistics	Cold/off-peak	50,000	Yes	6	5	0.75	524.3
JD Logistics	Cold/off-peak	50,000	Yes	5	5	0.75	476.6
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	428.9
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	381.1
JD Logistics	Cold/off-peak	50,000	No	9	6	0.71	759.3
JD Logistics	Cold/off-peak	50,000	Yes	8	5	0.75	659.0
JD Logistics	Cold/off-peak	50,000	Yes	7	5	0.75	606.4
JD Logistics	Cold/off-peak	50,000	Yes	6	5	0.75	553.8
JD Logistics	Cold/off-peak	50,000	Yes	5	5	0.75	501.1
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	448.5
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	395.8
JD Logistics	Cold/off-peak	50,000	No	9	6	0.71	790.1
JD Logistics	Cold/off-peak	50,000	No	8	5	0.75	686.5
JD Logistics	Cold/off-peak	50,000	Yes	7	5	0.75	630.4
JD Logistics	Cold/off-peak	50,000	Yes	6	5	0.75	574.4
JD Logistics	Cold/off-peak	50,000	Yes	5	5	0.75	518.3
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	462.2

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Brand	Season	Demand/day	3-year payback feasible?	Optimal robots	Optimal humans	Robot share	Daily cost ($)
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	406.1
JD Logistics	Cold/off-peak	50,000	No	9	6	0.71	847.5
JD Logistics	Cold/off-peak	50,000	No	8	5	0.75	737.5
JD Logistics	Cold/off-peak	50,000	No	7	5	0.75	675.0
JD Logistics	Cold/off-peak	50,000	Yes	6	5	0.75	612.6
JD Logistics	Cold/off-peak	50,000	Yes	5	5	0.75	550.2
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	487.7
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	425.3
JD Logistics	Cold/off-peak	50,000	No	1	18	0.0786	928.8
JD Logistics	Cold/off-peak	50,000	No	8	5	0.75	815.9
JD Logistics	Cold/off-peak	50,000	No	7	5	0.75	743.7
JD Logistics	Cold/off-peak	50,000	No	6	5	0.75	671.4
JD Logistics	Cold/off-peak	50,000	Yes	5	5	0.75	599.2
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	526.9
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	454.7
JD Logistics	Cold/off-peak	50,000	No	1	18	0.0786	948.4
JD Logistics	Cold/off-peak	50,000	No	4	12	0.38	938.5
JD Logistics	Cold/off-peak	50,000	No	7	5	0.75	881.0
JD Logistics	Cold/off-peak	50,000	No	6	5	0.75	789.1
JD Logistics	Cold/off-peak	50,000	No	5	5	0.75	697.3
JD Logistics	Cold/off-peak	50,000	Yes	4	5	0.75	605.4
JD Logistics	Cold/off-peak	50,000	Yes	3	5	0.75	513.5
JD Logistics	Normal	100,000	Yes	18	13	0.70	1,510
JD Logistics	Normal	100,000	Yes	15	13	0.70	1,367
JD Logistics	Normal	100,000	Yes	13	13	0.70	1,272
JD Logistics	Normal	100,000	Yes	11	13	0.69	1,176
JD Logistics	Normal	100,000	Yes	9	13	0.70	1,081
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,033
JD Logistics	Normal	100,000	Yes	6	13	0.70	937.6
JD Logistics	Normal	100,000	No	17	14	0.67	1,596
JD Logistics	Normal	100,000	Yes	15	13	0.70	1,441
JD Logistics	Normal	100,000	Yes	13	13	0.70	1,335
JD Logistics	Normal	100,000	Yes	11	13	0.69	1,230
JD Logistics	Normal	100,000	Yes	9	13	0.70	1,125
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,072
JD Logistics	Normal	100,000	Yes	6	13	0.70	967.0
JD Logistics	Normal	100,000	No	17	14	0.67	1,654
JD Logistics	Normal	100,000	Yes	15	13	0.70	1,492
JD Logistics	Normal	100,000	Yes	13	13	0.70	1,380
JD Logistics	Normal	100,000	Yes	11	13	0.69	1,268
JD Logistics	Normal	100,000	Yes	9	13	0.70	1,156
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,100
JD Logistics	Normal	100,000	Yes	6	13	0.70	987.6
JD Logistics	Normal	100,000	No	17	14	0.67	1,763
JD Logistics	Normal	100,000	No	15	13	0.70	1,588
JD Logistics	Normal	100,000	Yes	13	13	0.70	1,463
JD Logistics	Normal	100,000	Yes	11	13	0.69	1,338
JD Logistics	Normal	100,000	Yes	9	13	0.70	1,213
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,151
JD Logistics	Normal	100,000	Yes	6	13	0.70	1,026
JD Logistics	Normal	100,000	No	17	14	0.67	1,930
JD Logistics	Normal	100,000	No	15	13	0.70	1,735
JD Logistics	Normal	100,000	No	13	13	0.70	1,590
JD Logistics	Normal	100,000	No	11	13	0.69	1,446
JD Logistics	Normal	100,000	Yes	9	13	0.70	1,301
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,229
JD Logistics	Normal	100,000	Yes	6	13	0.70	1,085
JD Logistics	Normal	100,000	No	0	42	0	2,104
JD Logistics	Normal	100,000	No	15	13	0.70	2,029
JD Logistics	Normal	100,000	No	13	13	0.70	1,845
JD Logistics	Normal	100,000	No	11	13	0.69	1,662
JD Logistics	Normal	100,000	No	9	13	0.70	1,478
JD Logistics	Normal	100,000	Yes	8	13	0.70	1,386
JD Logistics	Normal	100,000	Yes	6	13	0.70	1,202
JD Logistics	Hot/peak	200,000	Yes	32	40	0.60	3,932
JD Logistics	Hot/peak	200,000	Yes	27	40	0.60	3,693
JD Logistics	Hot/peak	200,000	Yes	23	40	0.60	3,502
JD Logistics	Hot/peak	200,000	Yes	20	40	0.60	3,359
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,168
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,073
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	2,882
JD Logistics	Hot/peak	200,000	Yes	32	40	0.60	4,089
JD Logistics	Hot/peak	200,000	Yes	27	40	0.60	3,826
JD Logistics	Hot/peak	200,000	Yes	23	40	0.60	3,615

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Brand	Season	Demand/day	3-year payback feasible?	Optimal robots	Optimal humans	Robot share	Daily cost ($)
JD Logistics	Hot/peak	200,000	Yes	20	40	0.60	3,457
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,247
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,141
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	2,931
JD Logistics	Hot/peak	200,000	Yes	32	40	0.60	4,199
JD Logistics	Hot/peak	200,000	Yes	27	40	0.60	3,918
JD Logistics	Hot/peak	200,000	Yes	23	40	0.60	3,694
JD Logistics	Hot/peak	200,000	Yes	20	40	0.60	3,526
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,302
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,189
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	2,965
JD Logistics	Hot/peak	200,000	No	32	40	0.60	4,403
JD Logistics	Hot/peak	200,000	Yes	27	40	0.60	4,090
JD Logistics	Hot/peak	200,000	Yes	23	40	0.60	3,841
JD Logistics	Hot/peak	200,000	Yes	20	40	0.60	3,653
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,404
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,279
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	3,029
JD Logistics	Hot/peak	200,000	No	32	40	0.60	4,716
JD Logistics	Hot/peak	200,000	No	27	40	0.60	4,355
JD Logistics	Hot/peak	200,000	No	23	40	0.60	4,066
JD Logistics	Hot/peak	200,000	Yes	20	40	0.60	3,849
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,560
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,416
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	3,127
JD Logistics	Hot/peak	200,000	No	32	40	0.60	5,344
JD Logistics	Hot/peak	200,000	No	27	40	0.60	4,885
JD Logistics	Hot/peak	200,000	No	23	40	0.60	4,517
JD Logistics	Hot/peak	200,000	No	20	40	0.60	4,242
JD Logistics	Hot/peak	200,000	Yes	16	40	0.60	3,874
JD Logistics	Hot/peak	200,000	Yes	14	40	0.60	3,691
JD Logistics	Hot/peak	200,000	Yes	10	40	0.60	3,323

Table E10: Simulation_Grid - column block 3

Continuation of Simulation_Grid; first columns are repeated as row identifiers.

Brand	Season	Demand/day	All-human cost ($)	Savings %
SF Express	Cold/off-peak	50,000	675.7	0.0508
SF Express	Cold/off-peak	50,000	675.7	0.17
SF Express	Cold/off-peak	50,000	675.7	0.24
SF Express	Cold/off-peak	50,000	675.7	0.31
SF Express	Cold/off-peak	50,000	675.7	0.38
SF Express	Cold/off-peak	50,000	675.7	0.45
SF Express	Cold/off-peak	50,000	675.7	0.52
SF Express	Cold/off-peak	50,000	675.7	0.0217
SF Express	Cold/off-peak	50,000	675.7	0.12
SF Express	Cold/off-peak	50,000	675.7	0.20
SF Express	Cold/off-peak	50,000	675.7	0.27
SF Express	Cold/off-peak	50,000	675.7	0.34
SF Express	Cold/off-peak	50,000	675.7	0.42
SF Express	Cold/off-peak	50,000	675.7	0.50
SF Express	Cold/off-peak	50,000	675.7	0.0014
SF Express	Cold/off-peak	50,000	675.7	0.0858
SF Express	Cold/off-peak	50,000	675.7	0.17
SF Express	Cold/off-peak	50,000	675.7	0.24
SF Express	Cold/off-peak	50,000	675.7	0.32
SF Express	Cold/off-peak	50,000	675.7	0.40
SF Express	Cold/off-peak	50,000	675.7	0.48
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0.0198
SF Express	Cold/off-peak	50,000	675.7	0.11
SF Express	Cold/off-peak	50,000	675.7	0.18
SF Express	Cold/off-peak	50,000	675.7	0.27
SF Express	Cold/off-peak	50,000	675.7	0.36
SF Express	Cold/off-peak	50,000	675.7	0.46
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0.0264
SF Express	Cold/off-peak	50,000	675.7	0.0918
SF Express	Cold/off-peak	50,000	675.7	0.20
SF Express	Cold/off-peak	50,000	675.7	0.31
SF Express	Cold/off-peak	50,000	675.7	0.41
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0
SF Express	Cold/off-peak	50,000	675.7	0.0614
SF Express	Cold/off-peak	50,000	675.7	0.19
SF Express	Cold/off-peak	50,000	675.7	0.33
SF Express	Normal	100,000	1,517	0.12
SF Express	Normal	100,000	1,517	0.22
SF Express	Normal	100,000	1,517	0.28
SF Express	Normal	100,000	1,517	0.34
SF Express	Normal	100,000	1,517	0.40
SF Express	Normal	100,000	1,517	0.44
SF Express	Normal	100,000	1,517	0.50
SF Express	Normal	100,000	1,517	0.0700
SF Express	Normal	100,000	1,517	0.17
SF Express	Normal	100,000	1,517	0.24
SF Express	Normal	100,000	1,517	0.31

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Brand	Season	Demand/day	All-human cost ($)	Savings %
SF Express	Normal	100,000	1,517	0.38
SF Express	Normal	100,000	1,517	0.41
SF Express	Normal	100,000	1,517	0.48
SF Express	Normal	100,000	1,517	0.0338
SF Express	Normal	100,000	1,517	0.14
SF Express	Normal	100,000	1,517	0.21
SF Express	Normal	100,000	1,517	0.28
SF Express	Normal	100,000	1,517	0.35
SF Express	Normal	100,000	1,517	0.40
SF Express	Normal	100,000	1,517	0.47
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0.0801
SF Express	Normal	100,000	1,517	0.16
SF Express	Normal	100,000	1,517	0.23
SF Express	Normal	100,000	1,517	0.32
SF Express	Normal	100,000	1,517	0.37
SF Express	Normal	100,000	1,517	0.44
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0.0848
SF Express	Normal	100,000	1,517	0.16
SF Express	Normal	100,000	1,517	0.26
SF Express	Normal	100,000	1,517	0.32
SF Express	Normal	100,000	1,517	0.40
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0
SF Express	Normal	100,000	1,517	0.0215
SF Express	Normal	100,000	1,517	0.14
SF Express	Normal	100,000	1,517	0.23
SF Express	Normal	100,000	1,517	0.32
SF Express	Hot/peak	200,000	4,324	0.24
SF Express	Hot/peak	200,000	4,324	0.29
SF Express	Hot/peak	200,000	4,324	0.34
SF Express	Hot/peak	200,000	4,324	0.37
SF Express	Hot/peak	200,000	4,324	0.42
SF Express	Hot/peak	200,000	4,324	0.44
SF Express	Hot/peak	200,000	4,324	0.48
SF Express	Hot/peak	200,000	4,324	0.20
SF Express	Hot/peak	200,000	4,324	0.26
SF Express	Hot/peak	200,000	4,324	0.31
SF Express	Hot/peak	200,000	4,324	0.35
SF Express	Hot/peak	200,000	4,324	0.40
SF Express	Hot/peak	200,000	4,324	0.42
SF Express	Hot/peak	200,000	4,324	0.47
SF Express	Hot/peak	200,000	4,324	0.18
SF Express	Hot/peak	200,000	4,324	0.24
SF Express	Hot/peak	200,000	4,324	0.29
SF Express	Hot/peak	200,000	4,324	0.33
SF Express	Hot/peak	200,000	4,324	0.38
SF Express	Hot/peak	200,000	4,324	0.41
SF Express	Hot/peak	200,000	4,324	0.46
SF Express	Hot/peak	200,000	4,324	0.13
SF Express	Hot/peak	200,000	4,324	0.20
SF Express	Hot/peak	200,000	4,324	0.26

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Brand	Season	Demand/day	All-human cost ($)	Savings %
SF Express	Hot/peak	200,000	4,324	0.30
SF Express	Hot/peak	200,000	4,324	0.36
SF Express	Hot/peak	200,000	4,324	0.39
SF Express	Hot/peak	200,000	4,324	0.45
SF Express	Hot/peak	200,000	4,324	0.0610
SF Express	Hot/peak	200,000	4,324	0.14
SF Express	Hot/peak	200,000	4,324	0.21
SF Express	Hot/peak	200,000	4,324	0.26
SF Express	Hot/peak	200,000	4,324	0.32
SF Express	Hot/peak	200,000	4,324	0.36
SF Express	Hot/peak	200,000	4,324	0.43
SF Express	Hot/peak	200,000	4,324	0.0051
SF Express	Hot/peak	200,000	4,324	0.0296
SF Express	Hot/peak	200,000	4,324	0.11
SF Express	Hot/peak	200,000	4,324	0.17
SF Express	Hot/peak	200,000	4,324	0.25
SF Express	Hot/peak	200,000	4,324	0.30
SF Express	Hot/peak	200,000	4,324	0.38
JD Logistics	Cold/off-peak	50,000	951.8	0.25
JD Logistics	Cold/off-peak	50,000	951.8	0.35
JD Logistics	Cold/off-peak	50,000	951.8	0.40
JD Logistics	Cold/off-peak	50,000	951.8	0.45
JD Logistics	Cold/off-peak	50,000	951.8	0.50
JD Logistics	Cold/off-peak	50,000	951.8	0.55
JD Logistics	Cold/off-peak	50,000	951.8	0.60
JD Logistics	Cold/off-peak	50,000	951.8	0.20
JD Logistics	Cold/off-peak	50,000	951.8	0.31
JD Logistics	Cold/off-peak	50,000	951.8	0.36
JD Logistics	Cold/off-peak	50,000	951.8	0.42
JD Logistics	Cold/off-peak	50,000	951.8	0.47
JD Logistics	Cold/off-peak	50,000	951.8	0.53
JD Logistics	Cold/off-peak	50,000	951.8	0.58
JD Logistics	Cold/off-peak	50,000	951.8	0.17
JD Logistics	Cold/off-peak	50,000	951.8	0.28
JD Logistics	Cold/off-peak	50,000	951.8	0.34
JD Logistics	Cold/off-peak	50,000	951.8	0.40
JD Logistics	Cold/off-peak	50,000	951.8	0.46
JD Logistics	Cold/off-peak	50,000	951.8	0.51
JD Logistics	Cold/off-peak	50,000	951.8	0.57
JD Logistics	Cold/off-peak	50,000	951.8	0.11
JD Logistics	Cold/off-peak	50,000	951.8	0.23
JD Logistics	Cold/off-peak	50,000	951.8	0.29
JD Logistics	Cold/off-peak	50,000	951.8	0.36
JD Logistics	Cold/off-peak	50,000	951.8	0.42
JD Logistics	Cold/off-peak	50,000	951.8	0.49
JD Logistics	Cold/off-peak	50,000	951.8	0.55
JD Logistics	Cold/off-peak	50,000	951.8	0.0241
JD Logistics	Cold/off-peak	50,000	951.8	0.14
JD Logistics	Cold/off-peak	50,000	951.8	0.22
JD Logistics	Cold/off-peak	50,000	951.8	0.29
JD Logistics	Cold/off-peak	50,000	951.8	0.37
JD Logistics	Cold/off-peak	50,000	951.8	0.45
JD Logistics	Cold/off-peak	50,000	951.8	0.52
JD Logistics	Cold/off-peak	50,000	951.8	0.0035
JD Logistics	Cold/off-peak	50,000	951.8	0.0139

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Brand	Season	Demand/day	All-human cost ($)	Savings %
JD Logistics	Cold/off-peak	50,000	951.8	0.0743
JD Logistics	Cold/off-peak	50,000	951.8	0.17
JD Logistics	Cold/off-peak	50,000	951.8	0.27
JD Logistics	Cold/off-peak	50,000	951.8	0.36
JD Logistics	Cold/off-peak	50,000	951.8	0.46
JD Logistics	Normal	100,000	2,104	0.28
JD Logistics	Normal	100,000	2,104	0.35
JD Logistics	Normal	100,000	2,104	0.40
JD Logistics	Normal	100,000	2,104	0.44
JD Logistics	Normal	100,000	2,104	0.49
JD Logistics	Normal	100,000	2,104	0.51
JD Logistics	Normal	100,000	2,104	0.55
JD Logistics	Normal	100,000	2,104	0.24
JD Logistics	Normal	100,000	2,104	0.32
JD Logistics	Normal	100,000	2,104	0.37
JD Logistics	Normal	100,000	2,104	0.42
JD Logistics	Normal	100,000	2,104	0.47
JD Logistics	Normal	100,000	2,104	0.49
JD Logistics	Normal	100,000	2,104	0.54
JD Logistics	Normal	100,000	2,104	0.21
JD Logistics	Normal	100,000	2,104	0.29
JD Logistics	Normal	100,000	2,104	0.34
JD Logistics	Normal	100,000	2,104	0.40
JD Logistics	Normal	100,000	2,104	0.45
JD Logistics	Normal	100,000	2,104	0.48
JD Logistics	Normal	100,000	2,104	0.53
JD Logistics	Normal	100,000	2,104	0.16
JD Logistics	Normal	100,000	2,104	0.25
JD Logistics	Normal	100,000	2,104	0.30
JD Logistics	Normal	100,000	2,104	0.36
JD Logistics	Normal	100,000	2,104	0.42
JD Logistics	Normal	100,000	2,104	0.45
JD Logistics	Normal	100,000	2,104	0.51
JD Logistics	Normal	100,000	2,104	0.0829
JD Logistics	Normal	100,000	2,104	0.18
JD Logistics	Normal	100,000	2,104	0.24
JD Logistics	Normal	100,000	2,104	0.31
JD Logistics	Normal	100,000	2,104	0.38
JD Logistics	Normal	100,000	2,104	0.42
JD Logistics	Normal	100,000	2,104	0.48
JD Logistics	Normal	100,000	2,104	0
JD Logistics	Normal	100,000	2,104	0.0355
JD Logistics	Normal	100,000	2,104	0.12
JD Logistics	Normal	100,000	2,104	0.21
JD Logistics	Normal	100,000	2,104	0.30
JD Logistics	Normal	100,000	2,104	0.34
JD Logistics	Normal	100,000	2,104	0.43
JD Logistics	Hot/peak	200,000	5,951	0.34
JD Logistics	Hot/peak	200,000	5,951	0.38
JD Logistics	Hot/peak	200,000	5,951	0.41
JD Logistics	Hot/peak	200,000	5,951	0.44
JD Logistics	Hot/peak	200,000	5,951	0.47
JD Logistics	Hot/peak	200,000	5,951	0.48
JD Logistics	Hot/peak	200,000	5,951	0.52
JD Logistics	Hot/peak	200,000	5,951	0.31

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Brand	Season	Demand/day	All-human cost ($)	Savings %
JD Logistics	Hot/peak	200,000	5,951	0.36
JD Logistics	Hot/peak	200,000	5,951	0.39
JD Logistics	Hot/peak	200,000	5,951	0.42
JD Logistics	Hot/peak	200,000	5,951	0.45
JD Logistics	Hot/peak	200,000	5,951	0.47
JD Logistics	Hot/peak	200,000	5,951	0.51
JD Logistics	Hot/peak	200,000	5,951	0.29
JD Logistics	Hot/peak	200,000	5,951	0.34
JD Logistics	Hot/peak	200,000	5,951	0.38
JD Logistics	Hot/peak	200,000	5,951	0.41
JD Logistics	Hot/peak	200,000	5,951	0.45
JD Logistics	Hot/peak	200,000	5,951	0.46
JD Logistics	Hot/peak	200,000	5,951	0.50
JD Logistics	Hot/peak	200,000	5,951	0.26
JD Logistics	Hot/peak	200,000	5,951	0.31
JD Logistics	Hot/peak	200,000	5,951	0.35
JD Logistics	Hot/peak	200,000	5,951	0.39
JD Logistics	Hot/peak	200,000	5,951	0.43
JD Logistics	Hot/peak	200,000	5,951	0.45
JD Logistics	Hot/peak	200,000	5,951	0.49
JD Logistics	Hot/peak	200,000	5,951	0.21
JD Logistics	Hot/peak	200,000	5,951	0.27
JD Logistics	Hot/peak	200,000	5,951	0.32
JD Logistics	Hot/peak	200,000	5,951	0.35
JD Logistics	Hot/peak	200,000	5,951	0.40
JD Logistics	Hot/peak	200,000	5,951	0.43
JD Logistics	Hot/peak	200,000	5,951	0.47
JD Logistics	Hot/peak	200,000	5,951	0.10
JD Logistics	Hot/peak	200,000	5,951	0.18
JD Logistics	Hot/peak	200,000	5,951	0.24
JD Logistics	Hot/peak	200,000	5,951	0.29
JD Logistics	Hot/peak	200,000	5,951	0.35
JD Logistics	Hot/peak	200,000	5,951	0.38
JD Logistics	Hot/peak	200,000	5,951	0.44

Optimization G1

Table E11: Optimization_G1 - column block 1

This table reports the optimal robot-human mix at different assumed robot throughputs.

Brand	Season	Demand/day	q (pph)	Robots	Humans	Robot processed	Human processed
SF Express	Cold/off-peak	50,000	300	7	5	33,007	16,993
SF Express	Cold/off-peak	50,000	400	6	4	37,500	12,500
SF Express	Cold/off-peak	50,000	500	5	4	37,500	12,500
SF Express	Normal	100,000	300	14	11	66,014	33,986
SF Express	Normal	100,000	400	11	10	69,158	30,842
SF Express	Normal	100,000	500	9	10	70,000	30,000
SF Express	Hot/peak	200,000	300	27	31	120,000	80,000
SF Express	Hot/peak	200,000	400	20	31	119,454	80,546
SF Express	Hot/peak	200,000	500	16	31	119,454	80,546
JD Logistics	Cold/off-peak	50,000	300	8	5	37,500	12,500
JD Logistics	Cold/off-peak	50,000	400	6	5	37,500	12,500
JD Logistics	Cold/off-peak	50,000	500	5	5	37,500	12,500
JD Logistics	Normal	100,000	300	15	13	70,000	30,000
JD Logistics	Normal	100,000	400	11	13	69,158	30,842
JD Logistics	Normal	100,000	500	9	13	70,000	30,000
JD Logistics	Hot/peak	200,000	300	27	40	120,000	80,000
JD Logistics	Hot/peak	200,000	400	20	40	119,454	80,546
JD Logistics	Hot/peak	200,000	500	16	40	119,454	80,546

Table E12: Optimization_G1 - column block 2

Continuation of Optimization_G1; first columns are repeated as row identifiers.

Brand	Season	Demand/day	Robot share	Daily hybrid cost	All-human workers	All-human cost	Savings %
SF Express	Cold/off-peak	50,000	0.66	617.7	15	675.7	0.0858
SF Express	Cold/off-peak	50,000	0.75	516.6	15	675.7	0.24
SF Express	Cold/off-peak	50,000	0.75	460.5	15	675.7	0.32
SF Express	Normal	100,000	0.66	1,307	32	1,517	0.14
SF Express	Normal	100,000	0.69	1,091	32	1,517	0.28
SF Express	Normal	100,000	0.70	978.8	32	1,517	0.35
SF Express	Hot/peak	200,000	0.60	3,278	76	4,324	0.24
SF Express	Hot/peak	200,000	0.60	2,885	76	4,324	0.33
SF Express	Hot/peak	200,000	0.60	2,661	76	4,324	0.38
JD Logistics	Cold/off-peak	50,000	0.75	686.5	20	951.8	0.28
JD Logistics	Cold/off-peak	50,000	0.75	574.4	20	951.8	0.40
JD Logistics	Cold/off-peak	50,000	0.75	518.3	20	951.8	0.46
JD Logistics	Normal	100,000	0.70	1,492	42	2,104	0.29
JD Logistics	Normal	100,000	0.69	1,268	42	2,104	0.40
JD Logistics	Normal	100,000	0.70	1,156	42	2,104	0.45
JD Logistics	Hot/peak	200,000	0.60	3,918	99	5,951	0.34
JD Logistics	Hot/peak	200,000	0.60	3,526	99	5,951	0.41
JD Logistics	Hot/peak	200,000	0.60	3,302	99	5,951	0.45

Model Formulas

Table E13: Model Formulas

Clean symbolic specification for paper writing.

Component	Formula	Meaning	Decision use
Robot annual economic cost	$C_R(P) = aP + b$	$a = 0.306$ and $b = 13{,}362.44$ under the baseline TCO assumptions.	Maps bare robot price to annual economic cost.
Robot annual output	$Q_R(q, s) = q\, h_R\, d\, u_R\, \theta_R\, (1 - r_R)\, \phi_s\, m_s$	Throughput multiplied by operating hours, workdays, uptime, handling quality, robotable parcel share and seasonal capacity.	Maps robot efficiency to annual good parcels.
Seasonal human unit cost	$UC_{H,js} = UC_{H,j} \dfrac{\lambda_s}{\eta_s}$	Labour cost multiplier divided by the human capacity multiplier.	Makes peak and off-peak labour costs comparable.
Unit-cost break-even	$\dfrac{C_R(P)}{Q_R(q, s)} \leq UC_{H,js}$	Robot unit cost must not exceed human unit cost.	Operational cost parity.
Required robot throughput	$q^{UC}_{js}(P) = \dfrac{C_R(P)}{UC_{H,js}\, h_R\, d\, u_R\, \theta_R\, (1 - r_R)\, \phi_s\, m_s}$	Minimum parcels per hour required for a robot with bare price P.	Answers how efficient the robot must be.
Maximum robot price	$P^{UC}_{js}(q) = \dfrac{UC_{H,js} Q_R(q, s) - b}{a}$	Maximum bare robot price allowed at throughput q under unit-cost parity.	Answers how cheap the robot must be.
Three-year payback frontier	$P^{PB}_{js}(q) = \dfrac{T\, UC_{H,js} Q_R(q, s) - T\, O_F - B}{(1 + \alpha_E + \alpha_S + \alpha_G) + T\alpha_M}$	Bare robot price ceiling under the cash payback target T.	Main investment threshold.
Integer optimisation objective	$\min\limits_{R,H} \left[R \dfrac{C_R(P)}{d} + H \dfrac{C_{H,j}\lambda_s}{d} \right]$	Chooses the number of robots R and human workers H that minimise daily operating cost.	Determines the best hybrid robot–human mix.
Capacity constraint	$\min\{RK_R(q, s),\, \phi_s D_s\} + HK_{H,js} \geq D_s$	Robots can only cover the robotable share of demand; humans cover residual demand and exceptions.	Prevents unrealistic full automation.

Sources

Table E14: Sources

All row-wise URLs are retained as plain text to support auditability.

Input block	Source	How used	Data quality	Limitation
Salary components	Uploaded BOSS/Liepin extraction table	Employer cost, salary components, night shift, formal-worker status, housing/meal/social-insurance indicators	Medium	Recruitment pages are dynamic; figures are public postings rather than internal payroll
Human efficiency	Uploaded efficiency and quality workbook	Human pph assumptions and quality-loss ranges; SF 500 pph benchmark; JD estimates where worker-level data are not public	Medium	Employee-level wrong sorting, missed scanning, damage and rework rates are mostly estimated
Robot prices and TCO	Uploaded humanoid robot 2026 price/BOM/TCO workbook	Robot prices, scanner, software, maintenance, downtime, operator allocation and other TCO items	Medium	Industrial contract prices and maintenance contracts are incomplete
Unitree G1 benchmark	Public robot price in previous robot workbook	Main reference robot with P=$13,500	High for public price; medium for logistics deployment	Public price does not guarantee industrial parcel-sorting readiness
Seasonal simulation assumptions	Model assumption sheet	Cold/off-peak, normal and hot/peak demand with labour and capacity multipliers	Scenario assumption	Site-level parcel volume must be replaced by real facility data when available